

### Homework 1.

- The file name of your homework (in PDF) should be in the format: “學號-ds-作業編號.pdf”. For example: **00757999-ds-hw1.pdf**
- Please submit your homework to Tronclass **before 23:59, October 19, 2025**

1. (10%) Order the following function by growth rate in increasing order:  
 $n^2$ ,  $n \log(\log n)$ ,  $n \log(n^2)$ ,  $n^n$

2. (10%) Ackerman's function  $A(m, n)$  is defined as:

$$A(m, n) = \begin{cases} n + 1, & \text{if } m = 0 \\ A(m - 1, 1), & \text{if } n = 0 \\ A(m - 1, A(m, n - 1)), & \text{otherwise} \end{cases}$$

What is the value of  $A(3, 4)$ ?

3. (12%) Show that the following statements are correct:

- (a)  $5n^2 - 6n = \Theta(n^2)$
- (b)  $n! = O(n^n)$

4. (12%) Show that the following statements are incorrect:

- (a)  $10n^2 + 9 = O(n)$
- (b)  $3^n = O(2^n)$

5. (16%) (from NTHU 106) Which ones of the following conjectures are correct?

Please justify your answers:

- (a)  $3n^2 - 100n + 6 = O(n)$
- (b)  $\log(n!) = \Theta(n \log n)$
- (c)  $n! = O(n^n)$
- (d)  $(n + a)^b = \Theta(n^b)$  where  $a$  and  $b$  are real constants
- (e)  $(n + a)^b = \Omega(n^{b+1})$  where  $a$  and  $b$  are real constants

6. (20%) Fill the step count table (yellow regions) for the following program:

| Statement                                                                 | s/e | Frequency | Total Steps |
|---------------------------------------------------------------------------|-----|-----------|-------------|
| <b>void mult(int a[][MAX_SIZE], int b[][MAX_SIZE], int c[][MAX_SIZE])</b> | 0   | 0         | 0           |
| <b>{</b>                                                                  | 0   | 0         | 0           |
| <b>int i, j, k;</b>                                                       | 0   | 0         | 0           |
| <b>for (i = 0; i &lt; MAX_SIZE; i++)</b>                                  | 1   |           |             |
| <b>for (j = 0; j &lt; MAX_SIZE; j++) {</b>                                | 1   |           |             |
| c [ i ][ j ] = 0;                                                         | 1   |           |             |
| <b>for (k = 0; k &lt; MAX_SIZE; k++)</b>                                  | 1   |           |             |
| c [ i ][ j ] += a [ i ][ k ] * b [ k ][ j ];                              | 1   |           |             |
| <b>}</b>                                                                  | 0   | 0         | 0           |
| <b>}</b>                                                                  | 0   | 0         | 0           |
| Total                                                                     |     |           |             |

7. (10%) Given an four-dimensional array A[100][200][300][400], if the address of A[3][20][32][45] is  $\alpha + X$ , what is the value of X ?

Note: (1)  $\alpha$  is the address of A[0][0][0][0]

(2) Only consider row major here.

8. (10%) (from NTCU 107) Consider the following function. Please derive the time complexity of function foo using recurrence relation. Please show your derivation step by step.

```

int i[n]; /* Assume that the values of all integers in i[n] have been initialized */
int foo(int a, int b, int c) {
    int d;
    if (a>b) return -1;
    d=(a+b)/2;
    if (c==i[d])
        return d;
    else if (c>i[d])
        return foo(d+1,b,c);
    else
        return foo(a, d-1,c);
}

```